

MKUFT Integrated Master Spine

Professional reading edition of the evolving public MKUFT canon

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0. Purpose and claim boundary

MKUFT is a speculative research framework, not an accepted completed physical theory.

This master spine is a **synthesis and routing document**. Dedicated modules own full derivations, definitions, experiments, references, and falsifiers. If a compressed expression here conflicts with its canonical owner, the canonical owner controls and this spine must be corrected.

MKUFT survives only where it remains:

- typed by domain and layer;
- compatible with established results in their demonstrated regimes;
- explicit about whether an equation is notation, scaffold, model, or derived mechanism;
- open to ordinary explanations and strong alternatives;
- falsifiable at the branch where the claim is made;
- reconstructable from several public entry routes without changing meaning.

1. Core research question

MKUFT asks whether stable outcomes can be studied through a recurring grammar of:

```
possibility
→ constraint
→ admissible transition
→ cost
→ coherence / stability
→ boundary
→ measurement / registration
→ falsification
→ outcome
```

The recurrence of this grammar across domains does not establish one identical mechanism across those domains.

2. Typed S–I–P–O architecture

```
S = Substrate / structured possibility
I = Information / relation / constraint
P = Physical expression and dynamics
O = Observer-positioned registration and bounded participation
```

The layers are typed addresses, not four ordinary spatial dimensions.

S — Substrate

A mathematical handle is the measure space

$$S = (\Omega, \Sigma, \mu).$$

This is a formal representation of a possibility/source domain. It is not evidence that a hidden material medium, lattice, or additional physical space has been detected.

For recursive live-state work, Module 33 distinguishes this ambient possibility scaffold from an effective possibility object S_t^{eff} that contains the currently reachable/addressable support. A realised update may change S_t^{eff} without implying that observation rewrites an ambient universal substrate.

I — Information

To avoid symbol collision, use

$$\mathcal{I} = L^2(\Omega, \mu), \quad i \in \mathcal{I},$$

where I remains the layer label, $\mathcal{I}$ the selected information-state/function space, and i one information structure.

Information earns independent physical significance only where a defined coupling adds measurable content beyond an adequate physical description.

P — Physical

The P-layer contains measurable physical states, fields, bodies, instruments, environments, timing, and dynamics. Physical claims require physical variables, units, controls, conservation accounting where applicable, and ordinary-physics baselines.

O — Observer

The O-layer contains observer-positioned registration, measurement context, record, attention, interpretation, and any bounded state variable that is operationally defined.

Observer language cannot replace a missing physical mechanism. Where measurement changes the physical system, that back-action remains P-layer physics; Module 33 represents the general measurement step as a typed $P \rightarrow (P, O)$ instrument.

3. Realisation scaffold

For event E , a working unnormalised weight is

$$\tilde{W}(E) = \int D_{\text{phys}}(E | i) W_{SI}(i | S, E) C_O(O | i, E) d\nu(i),$$

with

$$P_{\text{realized}}(E) = \frac{\tilde{W}(E)}{Z}.$$

D_{phys} represents the accepted physical contribution, W_{SI} a candidate substrate-to-information weighting, and C_O a bounded observer-condition term.

These are working scaffolds. Every additional term must be operationalised and the complete probability model must be normalisable.

The ordinary limit is mandatory:

$$P_{\text{realized}}(E) \approx P_{\text{phys}}(E)$$

when additional terms are absent, constant, negligible, or empirically unnecessary within a declared regime and tolerance.

Module 33 does not silently replace this event-weighting scaffold. It supplies the more general typed update architecture within which a branch-specific weighting, if required, must live.

4. Constrained traversal

Represent a state structure as

$$\mathcal{G} = (N, E_{\mathcal{G}}),$$

with trajectory

$$\gamma = (n_0 \rightarrow n_1 \rightarrow \dots \rightarrow n_k).$$

For one declared state space, a candidate traversal cost is

$$C[\gamma] = \int_{\gamma} \lambda(x) ds.$$

The path element, cost density, and resulting units must be defined.

A Gibbs-like path model requires a dimensionless exponent. With inverse cost scale β ,

$$P(B | A) = \frac{1}{Z_A} \sum_{\gamma \in \Gamma(A \rightarrow B)} \exp[-\beta C[\gamma]].$$

Alternatively, use an explicitly dimensionless normalised cost $\tilde{C}[\gamma]$ in $\exp[-\tilde{C}[\gamma]]$.

This is a candidate model, not a declaration that all probability is fundamentally path density.

5. Time-dependent traversal geometry

Where admissibility changes with time, use

$$\mathcal{G}(t) = (N, E_{\mathcal{G}}(t)).$$

A temporally addressed path may be represented as

$$\gamma_t = ((n_0, t_0) \rightarrow (n_1, t_1) \rightarrow \cdots \rightarrow (n_k, t_k)),$$

with

$$C[\gamma_t] = \int_{\gamma} \lambda(x(s), t(s)) ds.$$

A candidate conditional weighting is then

$$P(B, t_B \mid A, t_A) = \frac{1}{Z} \sum_{\gamma_t \in \Gamma_{t_A \rightarrow t_B}(A \rightarrow B)} \exp[-\beta C[\gamma_t]].$$

The stationary model is recovered when the relevant transition structure and cost are effectively time-independent over the interval.

Time dependence adds scientific value only where it improves prospective prediction beyond ordinary time-dependent models rather than adding retrospective flexibility.

6. Independent-content criterion

A central scientific distinction is whether an I-layer description is merely a useful representation of the adequate P-layer state or carries independently measurable predictive content.

Let:

- X_t = measured physical/environmental state;
- H_t = relevant measured physical history;
- I_t = candidate information-layer variable;
- $Y_{t+\Delta}$ = preregistered later physical outcome.

The physical-only null is schematically

$$H_0 : Y_{t+\Delta} \perp\!\!\!\perp I_t \mid X_t, H_t.$$

The stronger I-layer claim earns support only if I_t improves held-out prediction after X_t and H_t are controlled, survives appropriate relation-preserving re-encoding, and cannot be reduced to an adequate physical-state function.

If

$$I_t = f(X_t, H_t),$$

then the I-layer may remain a useful effective description or compression, but this test has not established independent I→P dynamics.

The same reduction discipline applies to observer variables.

7. Ambiguity dynamics

For a declared domain d , let $\Omega_t^{(d)}$ be the unresolved feasible region compatible with current evidence and constraints. A dimensionless ambiguity-volume index is

$$A_{t,\text{vol}}^{(d)} = \log\left(1 + \frac{\mu_d(\Omega_t^{(d)})}{\mu_{0,d}}\right).$$

With normalised route connectivity R_t and preserved access X_t ,

$$M_t = A_{t,\text{vol}}^{(d)} R_t X_t.$$

M_t is a heuristic audit index, not a universal law. The product form must compete with additive and interaction alternatives.

Canonical owner: [docs/21_AMBIGUITY_DYNAMICS_AND_MANOEUVRE_SPACE.md](#).

8. Cross-layer addressing

For candidate invariant K at layer L ,

$$K_L = A_L(K; \theta_L).$$

A cross-layer coupling must identify source and receiving spaces, variables, units or normalisation, observable consequence, ordinary baseline, and falsifier.

A typed coupling may be represented as

$$C_{LM} : \mathcal{X}_L \rightarrow \mathcal{X}_M.$$

Repeated algebraic form does not prove one physical mechanism.

The recursive integrity rule is:

No untyped inheritance.

Provenance, evidence, licence, causation, authority, responsibility, units, and other properties do not automatically transfer merely because two objects are linked.

Canonical owners: Modules 22 and 22A.

9. Recursive constraint closure and reachable-state geometry

For adaptive systems whose relevant variables, admissible operations, decoder, or update rule may change, Module 32 uses addressed state-space families rather than assuming that every update occurs inside one fixed coordinate system.

Let

$$\mathfrak{X} = \bigsqcup_{\alpha \in A} \{\alpha\} \times \mathcal{X}_\alpha.$$

A schematic adaptive update is

$$\mathfrak{F}_t(\mathfrak{A}_t, u_t) = (\alpha_{t+1}, x_{t+1}) \in \mathfrak{X}.$$

When $\alpha_{t+1} \neq \alpha_t$, direct subtraction or state equivalence requires a lawful translation, embedding, quotient, or common observable. Construction-address change is not silently treated as motion inside one fixed space.

The recursive candidate is:

```
previous organisation
→ consolidated typed constraints
→ changed admissibility
→ changed reachable states / capabilities
→ updated organisation
```

A scale-separated enabling-constraint pattern can occur when lower-scale feasible-state volume decreases while higher-scale viable capability increases:

$$\Delta\mu_\ell < 0, \quad \Delta\nu_L > 0,$$

provided each before/after difference is defined in a common declared comparison space. This is not a universal monotonic law. Overconstraint can reduce both quantities, and constraint count is not a proxy for coherence, agency, or freedom.

Recurrent architecture across domains supports a stronger comparison only when the recurrent **constraint relation and function** survive typed equivalence tests. Shape alone is insufficient.

9.1 Relational brackets, completion fibers, and scale realisation classes

Module 32S3 generalises the local-to-global object beyond literal spatial overlap. Let

$$\mathcal{L}_t = \prod_i \mathcal{F}_t(U_i)$$

be an ambient assignment space and

$$\mathfrak{E}_t = \{r_1, \dots, r_m\}$$

a typed family of relation scopes. Geometric overlap is one possible relation class; regulatory, contextual, system-environment, temporal, semantic/functional, and other relations require their own operational definitions. The product is a coordinate scaffold, not an independence assumption.

Typed compatibility tests select

$$\mathfrak{E}_{B,t} = \{a \in \mathcal{L}_t : \chi_B(a) = 1\}.$$

For a missing coordinate k , the compatible completion is the missing-coordinate image of the fiber over the retained assignment:

$$\begin{aligned} \mathfrak{F}_{k,t}(a_{-k}) &= \mathfrak{E}_{B,t} \cap \pi_{-k}^{-1}(a_{-k}), \\ \Omega_{k,t}^{\text{comp}}(a_{-k}) &= \pi_k(\mathfrak{F}_{k,t}(a_{-k})). \end{aligned}$$

This separates a narrow completion, broad ambiguity, local incompatibility, and unrelated background inconsistency. Relational specification does not imply physical occupancy.

A declared higher-scale readout

$$R_{\ell \rightarrow L} : \mathfrak{E}_{B,\ell} \supseteq \text{Dom}(R) \rightarrow \mathcal{Y}_L$$

induces equivalence classes. For macrostate y , the compatible lower-scale realisation class is

$$\boxed{\mathcal{M}_y = \mathfrak{E}_{B,\ell} \cap R_{\ell \rightarrow L}^{-1}(y)}.$$

Thus lower-scale state can change while the declared macrostate remains fixed whenever the trajectory remains inside $\mathcal{M}_y$. This is an operational model of lower-level turnover with higher-level persistence; it is not by itself a claim of personal identity or autonomous macro causation.

The quotient

$$\mathfrak{E}_{B,\ell} / \sim_R$$

is only a readout/coarse-graining until the Module 32S effective-object tests establish persistence, interventional load-bearingness, new viable capability, predictive compression, and boundary/closure specificity.

The scale handoff is schematically

```
ambient local possibilities
→ typed relation-compatible region
→ quotient/readout
```

- effective-object promotion when earned
- typed constituent in the next-scale bracket

with completion as the complementary zoom-in fiber operation.

Module 32S3 also defines the constraint-to-admissibility scaffold

$$\mathcal{Q}_{\lambda,t}(\mathbf{r}, x, h) = \Omega_{\lambda,t}^{\text{adm}}.$$

At the I→P boundary, a non-zero relational effect can remain fully P-realised. Independent I→P dynamics still require content beyond the strongest adequate P-state/history/boundary model.

9.2 Intrinsic/extrinsic address and transport

Module 32S4 separates intrinsic relational organisation from extrinsic/contextual address.

For intrinsic class κ , extrinsic address b and relevant history h ,

$$\mathcal{F}_P(\kappa; b, h) = \{p \in \mathfrak{P} : \chi_P(p; \kappa, b, h) = 1\}$$

is a context-indexed set of compatible P-realisations. Relational specification, compatible physical realisation and actual occupancy remain distinct.

For a contextual path $\gamma : b_0 \rightarrow b_1$, a lawful transport T_γ may be path-dependent. Same endpoints therefore do not imply identical transported state where a genuine connection/history dependence exists. Holonomy language is used only where its mathematical requirements are actually met.

Established boundary-conditioned field physics and metasurface holography are retained as P-layer comparators: context/boundary can alter physical mode structure or reconstructed gestalt, but this does not establish an independent I layer or free-energy source.

9.3 SIPO capstone — addressed law assembly

Module 33 composes the preceding dependencies into the live SIPO update.

Let the complete addressed state be

$$\mathcal{U}_t = (S_t^{\text{eff}}, I_t, P_t, O_t, b_t, H_t)$$

with measured/controlled context E_t where required. The typed admissibility descriptor is

$$\Xi_t = \mathcal{Q}_t(\mathcal{U}_t, E_t).$$

The active physical law object is

$$\mathfrak{L}_{P,t} = (\mathcal{D}_{P,t}, \mathcal{T}_{P,t}, \mathcal{W}_{P,t}),$$

where $\mathcal{D}$ is the addressed physical state/path/operator domain, $\mathcal{T}$ the physical transition rule/generator, and $\mathcal{W}$ any domain-appropriate weighting/selection object.

A typed law assembler maps

$$\mathfrak{C}_P : (\Xi_t, P_t, E_t) \mapsto \mathfrak{L}_{P,t}.$$

The minimal clean Layer-Before-Law branch is

$$\mathcal{D}_{P,t} = \mathfrak{D}_P(\Xi_t)$$

with standard $\mathcal{T}$ and $\mathcal{W}$ on that domain. The general capstone allows an independently demonstrated relation to affect $\mathcal{D}$, $\mathcal{T}$, $\mathcal{W}$, or a typed combination; domain-only conditioning is therefore a specialisation, not a universal restriction.

The SIPO update is

$$\text{Update}_{\text{SIPO}} = \text{Readdress} \circ \text{Instrument}_{P \rightarrow (P,O)} \circ \text{Propagate}_{\mathfrak{L}_P}^{\mathfrak{L}_P}, \quad \mathfrak{L}_P = \mathfrak{C}_P \circ \mathcal{Q} \text{ with typed state/context arguments.}$$

This means:

```
complete addressed state + context/history
→ typed admissibility descriptor  $\Xi$ 
→ assemble active P-law object (domain, transition, weighting)
→ propagate physically
→  $P \rightarrow (P, 0)$  instrument handles physical measurement back-action + record
→ realised path/history/context readdress the next effective possibility object
→ repeat
```

The capstone makes domain before generator and support before weighting explicit as clean logical specialisations without forcing all future $I \rightarrow P$ mechanisms into those forms. If the same law object is fully determined by an adequate P-state/history/boundary/environment model, the relation remains P-realised.

9.4 Dynamic interface promotion and recursive boundary closure

Module 33S1 refines the readdressing stage without changing the parent capstone. Let

$$\mathcal{J}_{AB,t}$$

be a typed interface/junction state between addressed systems A_t and B_t . An interface remains bundled context unless it is operationally definable/trackable, controlled deformation changes a preregistered load-bearing downstream object, and the realised transition changes the interface in a way that alters the next cycle.

The recursive signature is

$$\mathcal{J}_{AB,t} \rightarrow (A_{t+\Delta}, B_{t+\Delta}) \rightarrow \mathcal{J}_{AB,t+\Delta} \rightarrow \Xi_{t+\Delta}.$$

Passing this gate promotes the relation into explicit state representation; it does **not** establish independent ontology or an independent I layer. A fully P-recoverable interface may still be a useful macrostate or sufficient state variable.

If the coupled system later passes the existing Module 32S scale-promotion tests, the same lower-scale boundary/junction may become an internal relation of a promoted higher-scale whole. Physical work required to create, move, modulate or maintain an interface remains inside the energy ledger.

9.5 Relational closure and property-specific law descent

Module 33S2 separates existence of a higher-order functional object from dynamical sufficiency at that higher address.

Let

$$\pi_R : X \rightarrow Y$$

map lower-address states into candidate higher relational states under a declared load-bearing relation R . Functional equivalence at Y does not erase historical lineage, and promotion into Y does not automatically produce an autonomous law.

For matched deterministic lower dynamics F_X , strong whole-state descent requires

$$\pi_R(x) = \pi_R(x') \Rightarrow \pi_R(F_X(x)) = \pi_R(F_X(x'))$$

within the declared environment, intervention class, timescale and tolerance. Where only a target property q is claimed, descent is tested for q rather than silently certifying all of Y .

The supported class must be named as strong/exact, strong/approximate, ensemble/conditional, history-augmented, or coupled multiscale. Predictive sufficiency does not automatically grant interventional sufficiency.

Closure and law descent remain separate margins:

$$M_C$$

for loss of the constitutive closure relation, and

$$M_D(q)$$

for loss of higher-address sufficiency for property q . A regime with $M_D(q) < M_C$ is allowed: the organised whole can persist after a simpler higher-address law for q has failed.

Readdressing is therefore bidirectional:

lower address
 ↔ higher / effective address
 ↔ coupled multiscale address

with the active address earned by the current target property and evidence rather than inherited permanently from an earlier promotion.

9.6 Cross-scale performance and recoverability

Module 33S3 adds a further guard where a lower-address process changes performance while the enclosing organisation is also changing.

Let J_ℓ be a declared lower-address performance variable. Track it separately from closure and law descent:

$$\Delta_{\ell \rightarrow h} = (\Delta J_\ell, \Delta M_C, \Delta M_D(q)).$$

The tuple is retained rather than collapsed into one scalar because its entries may have different units and meanings. Cross-scale change may be aligned, antagonistic, neutral/decoupled, or mixed/property-relative. In particular,

$$\Delta J_\ell > 0, \quad \Delta M_C < 0$$

is a possible antagonistic pattern, not a universal law. Local improvement does not imply whole-level improvement, and it does not imply inevitable later collapse.

Recovery is defined conditionally. Let $\mathcal{C}_R \subseteq Y$ be a declared target closure class, with intervention/control class U , environment class E , admissible route constraints and horizon H . A state is recoverable only where an admissible route can reach $\mathcal{C}_R$ under those declared conditions. The notation

$$\text{Rec}_H(x_t; \mathcal{C}_R, U, E) = 1$$

is a reachability scaffold, not a universal recovery law.

Loss of a property-specific higher law, loss of closure, and loss of recoverability are therefore different events. A recovery route need not be the time reverse of the degradation route, and functional recovery may occur through a different lower-level realisation of the same target relational class.

Apparent hysteresis or path dependence must first be tested against bounded state augmentation. If adding a measurable omitted state or physical memory variable closes the process, that variable belongs in the address; history is not retained merely because it makes the account fit.

Canonical owners: Modules 22A, 32, 32S, 32S1, 32S2, 32S3, 32S4, 33, 33S1, 33S2, and 33S3.

10. LUCY

LUCY means **Local Unified Coherence Yield**.

Its canonical invariant is a thresholded addressed crossing:

addressed relational state
 → coherence / closure threshold
 → consequential change at the next addressed layer

Current formal work uses a yield index Y_L rather than overloading C :

$$Y_L(x, t) = \chi_L \frac{\|\nabla \tau_L(x, t)\|^2}{N_L(x, t) + \varepsilon_L},$$

with candidate threshold

$$Y_L(x, t) \geq Y_*.$$

Where a physical implementation produces boundary morphology:

LUCY-1 = membrane-like boundary condition, when present
LUCY-2 = sustained local region, when present

These are conditional downstream morphologies, not automatic consequences of every LUCY crossing.

Canonical owners: Modules 08, 22, 27, and the physical specialisation in Module 30. Modules 32S3/32S4/33/33S1/33S2/33S3 now supply increasingly general admissibility, transport, update, interface-promotion, law-descent and recoverability machinery around a candidate crossing; none of them by itself establishes independent $I \rightarrow P$ dynamics or a physical LUCY mechanism.

11. Observer and procedural support modules

OCQS

OCQS is a bounded observer-state hypothesis. It does not prove remote information, probability modulation, synchronicity, or substrate access. Its value depends on operational measures and comparison with established cognitive models.

GRACE

GRACE—Geometry, Relation, Admissibility, Coherence, Emergence—is a traversal filter. It does not create a new layer or supply evidence by acronym.

ESRT / ESF

ESRT/ESF test whether an artefact or representation contains reproducible **addressing, state, and flow** structure. A pass identifies an executable-system candidate, not a hidden meaning or historical use by assertion.

Voynich

The Voynich branch is an applied procedural hypothesis. It must compete with linguistic, cipher, scribal, decorative, mnemonic, diagrammatic, and null models under blinded, quantitative, held-out tests. It is not evidence for foundational MKUFT physics.

12. Foundational physics — Layer Before Law

The foundational proposal asks whether quantum and gravitational effective behaviour may arise from a deeper typed update architecture rather than being fundamental objects at the same descriptive layer.

The live-canon update is no longer only a placeholder arrow. Module 33 supplies the architectural factorisation above; Module 33S1 specifies when a load-bearing interface must be represented explicitly in the recursively readdressed state; Module 33S2 requires a promoted higher address to earn property-specific law sufficiency; and Module 33S3 prevents local performance or temporary degradation from being converted into unsupported whole-level stability, collapse, or recovery claims. The remaining burden is the **physical instantiation** of that architecture:

- instantiate Ξ_t and $\mathfrak{C}_P$ for each target regime;
- define the resulting $\mathcal{D}_{P,t}$, $\mathcal{T}_{P,t}$ and $\mathcal{W}_{P,t}$;
- derive Bell-compatible quantum correlations and no-signalling;
- recover the Born rule and relevant quantum/QFT limits rather than replacing them by assertion;
- recover general-relativistic or experimentally equivalent gravitational behaviour;
- preserve dimensional and conservation requirements;
- produce at least one operational difference from strong alternatives if the deeper architecture is to earn independent physical content.

If those burdens are not met, the capstone remains a coherent architectural factorisation rather than completed fundamental unification.

13. Experimental and promotion discipline

A broad interesting effect is not automatically an MKUFT result.

The public experimental route is:

```
calibration
→ reconnaissance
→ locked discriminating signature
→ strongest fair null
→ deformation / ablation
→ held-out prediction
→ independent replication
→ bounded interpretation
```

A branch is reduced or removed when its variables cannot be operationalised, simpler models predict equally or better, the effect disappears under controls, results do not replicate, definitions move after failure, or evidence from another layer is used to rescue it.

Canonical owners: Modules 04, 05, 27, 28, 29, and branch-specific controls in the 32/33 family.

14. Metaphysical boundary

MKUFT contains a wider metaphysical interpretation in which God and Love may have explicit philosophical roles. Those roles are not inserted into physical equations as unmeasured variables and cannot rescue a failed empirical claim.

Scientific claims remain answerable to the variables, measurements, alternatives, and falsifiers of the domain in which they are made.

15. Public integrity boundary

The public canon must not contain non-public personal data, private correspondence, credentials, internal-only operating instructions, or uncleared personal case material.

Development-stage context should be translated into a public invariant or retired rather than published merely because it helped discovery.

16. One-page compression

```
S-I-P-0:
Substrate → Information → Physical → Observer → next effective possibility

Information notation:
I = layer
J = selected information state/function space
i ∈ J = one information structure

Traversal:
G = (N, E, G)
C[y] = ∫ λ(x) ds
P(B|A) ∝ ∑ exp[-βC[y]]

Dynamic traversal:
G(t) = (N, E, G(t))
C[y_t] = ∫ λ(x(s), t(s)) ds

Independent-content null:
Y_(t+Δ) ⊥ I_t | X_t, H_t

Ambiguity:
A_t,vol = log(1 + μ(Q_t)/μ_0)
M_t = A_t,vol R_t X_t

Layer address:
K_L = A_L(K; θ_L)

Adaptive addressed space:
x = ∪_α ({α} × X_α)
x̂_t(u_t, u_t) = (α_(t+1), x_(t+1)) ∈ x

Relational bracket:
L_t = ∩_i F_t(U_i)
c_B,t = {a ∈ L_t : χ_B(a)=1}

Missing-address completion:
x̂_k,t(a_-k) = c_B,t ∩ π_-k^(-1)(a_-k)
```

```

Ωk,comp = πk( $\tilde{\gamma}_k$ ,t)

Macro realisation class:
 $\mathcal{M}_y = c_B, \ell \cap R_{-(t+L)}^{(-1)}(y)$ 

Context-indexed P realisation:
 $\mathcal{F}_P(\kappa;b,h) = \{p \in \mathbb{P} : \chi_P(p;\kappa,b,h)=1\}$ 

Complete addressed state:
 $\mathcal{U}_t = (S_t^{\text{eff}}, I_t, P_t, O_t, b_t, H_t)$ 

Admissibility descriptor:
 $\Xi_t = Q_t(\mathcal{U}_t, E_t)$ 

P-law object:
 $\mathcal{Q}_{P,t} = (\mathcal{D}_{P,t}, \mathcal{F}_{P,t}, \mathcal{W}_{P,t})$ 

Law assembly:
 $\mathcal{Q}_{P,t} = c_P(\Xi_t, P_t, E_t)$ 

Domain-only specialisation:
 $\mathcal{D}_{P,t} = \mathcal{D}_P(\Xi_t)$ 

SIPO capstone:
Update_SIPO = Readdress  $\circ$  Instrument_ $(P \sim (P,0))$   $\circ$  Propagate_ $P^{\wedge}(\mathcal{Q}_P)$ 

Dynamic interface promotion:
 $\mathcal{J}_{AB,t} \rightarrow (A_{-}(t+\Delta), B_{-}(t+\Delta)) \rightarrow \mathcal{J}_{AB,t+\Delta} \rightarrow \Xi_t + \Delta$ 
State promotion  $\neq$  ontology promotion

Relational law descent under matched conditions:
 $\pi_R(x) = \pi_R(x') \Rightarrow \pi_R(F_X(x)) = \pi_R(F_X(x'))$ 
Whole-state descent  $\neq$  property-specific descent for q
 $M_C \neq M_D(q)$ 

Cross-scale performance:
 $\Delta_{-(t+h)} = (\Delta \mathcal{J}_t, \Delta M_C, \Delta M_D(q))$ 
Local gain does not inherit upward

Conditional recovery:
RecH( $x_t; c_R, U, E$ ) = target-relative reachability under declared constraints
Recovery path need not invert degradation path

Enabling-constraint pair, when comparable:
 $\Delta \mu_t < 0$  while  $\Delta v_L > 0$ 

LUCY:
 $Y_L = \chi_L \ ||\nabla \tau_L||^2 / (N_L + \epsilon_L)$ 
 $Y_L \geq Y_{-*}$ 

Integrity:
Define once. Type the address. Build admissibility. Assemble the law object. Promote a relation into explicit state only when it carries prospective recursive load. Earn higher-address law ownership per property. Do not inherit local performance across scale. Treat recovery as conditional reachability. Test the coupling. Recover the baseline. Keep the falsifier.

```

17. Final statement

MKUFT should not read as a pile of similarities.

Its current live spine is a recursive research architecture in which **possibility, relation, admissibility, physical law, measurement/registration, history, interface state, scale transition, law descent and conditional recoverability** are explicitly separated but composable.

The architectural drivetrain remains closed by Module 33. Module 33S1 tightens recursive interface/state representation; Module 33S2 limits higher-address law ownership to the property and regime that actually earn it; Module 33S3 separates local performance from enclosing-scale viability and recovery from inevitable reversal. None reopens the parent capstone. The remaining foundational burden is not another missing arrow; it is to instantiate that drivetrain in concrete physics, recover the established quantum and gravitational regimes, and determine whether the deeper relational architecture earns independent predictive content.